Deep Learning Projects

Introduction

The goal of these projects is to design a deep learning methodology to solve a task on a provided dataset. This involves:

- Implementing methods to load and preprocess the data
- Implementing and training several relevant models and/or one model with several parameter sets
- Computing relevant metrics and analyzing the results
- Visualizing results

Choose one of the projects proposed in the next sections.

Specific expected outcomes are defined for each project, and the following criteria are used for **evaluation**:

Criterion	Explanation	Points
Method	- Relevant choice of methods - Several methods are compared - Decisions are justified	5
Reproducibility, code	- Provided code can be run easily - A readme file with instructions is provided	2
Evaluation of results	- Several relevant metrics are computed over train / val splits - Computational complexity (e.g. training time, inference time) is assessed	3
Report	- Report is clearly written - Graphics are readable and complete (e.g. axis titles) - Results are analysed - Discussion in terms of scientific (domain) output - Limitations are clearly discussed	2
TOTAL		**12**

Table 1: Evaluation criteria

1 Monitoring the evolution of glacier surface from SAR imagery

Background. Monitoring lake- and maritime-terminating glacier is important for hydrological and climatological studies. While glaciers regularly shrink and extend, rising temperatures lead to glacier calving and increased release of fresh water into the oceans. Synthetic Aperture Radar (SAR) is commonly used to monitor glaciers because this active sensing technology can penetrate clouds and obtain images during the polar nights. It creates images by measuring the relative backscattered energy (among other properties) from radar electromagnetic waves emitted and received by a moving antenna.

Task. The objective of this project is to measure the surface area of two ocean-terminating glaciers, `Columbia` and `Mapple`, over time. For this, you will train glacier segmentation algorithms on a training set of different glaciers, and use the predicted segmentation maps to measure glacier surface area for the test glaciers.
Predicting on unseen glaciers is challenging because of the large geographical domain gap and limited training diversity. With a method of your choice, you will use the first year of data for each test glacier to improve local predictions.

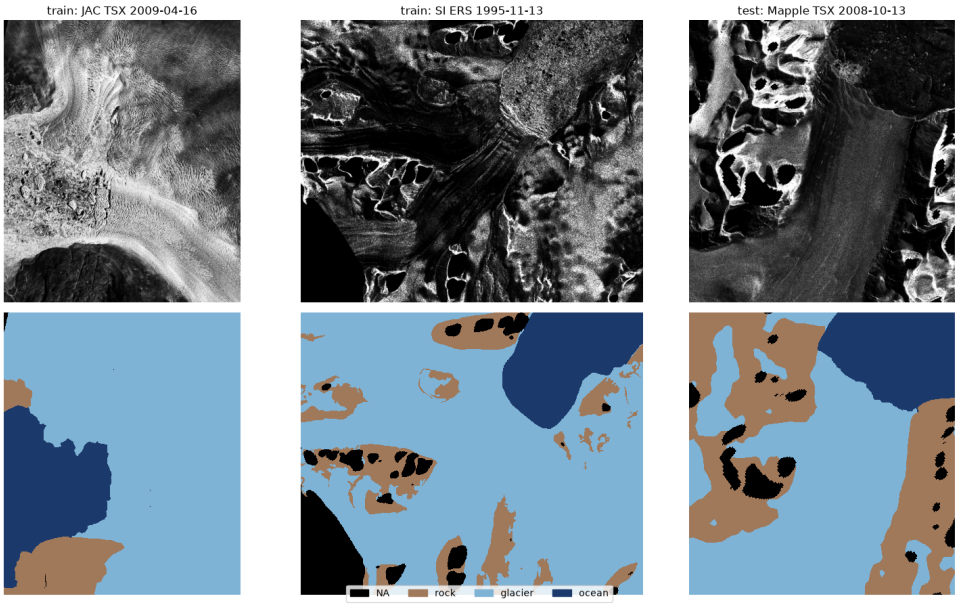


Figure 1: Examples of SAR images of glaciers from the train and test sets, with their corresponding segmentation maps.

Data. The CaFFe dataset [1] is composed of SAR images of seven lake and maritime terminating glaciers from Antarctica, Greenland and Alaska acquired between 1995 and 2020. The labels provided correspond to the segmentation map of multiple landscape elements: `ocean`, `rock outcrops`, `glacier area`, and `no information available` (often due to SAR shadowing). Data were acquired from six different sensors, each with their own specific resolution and characteristics.
We preprocessed the data from the original publication by down-sampling the images and standardizing the spatial resolution to 20m. For the test glaciers, we also provide a mask as the complement of the union of the "N/A" regions on which you can measure glacier extent. Otherwise, your results will vary depending on the amount of SAR shadows in each test image. Two test scenes were dropped as they did not fully cover the region. Segmentation labels were preprocessed similarly. The final dataset comprises 678 images (558 train, 120 test) at 20m resolution, of varying size (from 382×405 to about 1492×702 pixels).
The label encoding is: 0 = NA, 64 = rock, 127 = glacier, 254 = ocean You are free to use a validation set of your choice (from the training images) to monitor the performance of your segmentation model during training.

Data can be download here.

Expected Outcomes:

1. A first deep-learning based glacier segmentation method evaluated on the test set of images using mean Intersection-over-Union (mIoU) for the glacier class.
2. Exploration of one or both of the following*:
   (a) An extension of your model that makes use of the first year of data of each test glacier to improve glacier specificity (*for example* by fine-tuning it on this new set of data). We consider the first year of data as the set of test images that are acquired within 365 days after the first acquisition for a given glacier.
   (b) An extension of your model trained with an additional sensor discriminative loss that will improve sensor-specific predictions (*for example* by predicting the sensor type as an additional output).
3. Compare 1. and 2. on the test dataset. If you used the first year of test data to learn the model parameters (2.a), you must evaluate on the test set occurring after this first year.
4. For one of the implemented segmentation models of your choice, a quantification of the `Columbia` and `Mapple` glacier surface area over the years and a measure of the evolution trends with uncertainty. Discuss the relative error with respect to the ground-truth segmentation maps and per-sensor errors. We expect you to clearly evaluate if any of your implemented solutions is reliable for measuring glacier evolution over time. A well-argued negative result is totally acceptable.

* *If time allows, we encourage you to try other methods than the given examples. Developing both approaches is not necessary to get full points but will be considered as a bonus.*

References. [1]. Gourmelon, N., Seehaus, T., Braun, M., Maier, A. & Christlein, V. (2022). Calving fronts and where to find them: a benchmark dataset and methodology for automatic glacier calving front extraction from synthetic aperture radar imagery. Earth System Science Data, 14(9), 4287-4313.

2 Above-ground forest biomass estimation from multisensor satellite imagery

Task. Accurate estimates of above-ground forest biomass are important for understanding forest carbon stocks and monitoring ecosystem changes. Direct biomass measurements from field surveys and airborne LiDAR provide high-quality estimates but are costly to acquire over large areas. Satellite observations provide an opportunity to estimate biomass continuously over large regions.
In this project, you will develop deep learning models to estimate above-ground biomass from Sentinel-1 SAR and Sentinel-2 multispectral imagery. You will investigate the contribution and complementarity of optical and radar observations, explore methods for combining both data sources, and analyze which input channels or groups of channels provide the most useful information for biomass estimation. The goal is not only to obtain accurate biomass estimates, but also to understand which parts of the multisensor input drive model performance and whether these contributions are consistent with the physical information measured by the different sensors.

Data. Download the dataset here. The data are extracted from the BioMassters dataset [1] and cover forested areas in Finland. Each sample corresponds to a 2.56×2.56 km area represented on a 256×256 grid at 10 m spatial resolution. For each sample, we provide Sentinel-1 and Sentinel-2 observations along with a reference above-ground biomass map derived from airborne LiDAR and field measurements.

- **Images.** The Sentinel-1 data contain four SAR channels combining two orbit directions, ascending (ASC) and descending (DSC), with two radar polarizations, VV and VH: ASC VV, ASC VH, DSC VV, and DSC VH. The Sentinel-2 data contain the spectral bands B2, B3, B4, B5, B6, B7, B8, B8A, B11, and B12, covering the visible, red-edge, near-infrared, and short-wave infrared regions of the spectrum. A cloud probability layer (CLP) is also provided. The different channels therefore contain complementary information that can be analyzed individually when interpreting the trained models. All input images are spatially aligned.
- **Labels.** The target is a continuous above-ground biomass map at 10 m spatial resolution, aligned with the Sentinel-1 and Sentinel-2 inputs. Each pixel therefore corresponds to a biomass value for the same spatial location in the satellite imagery.

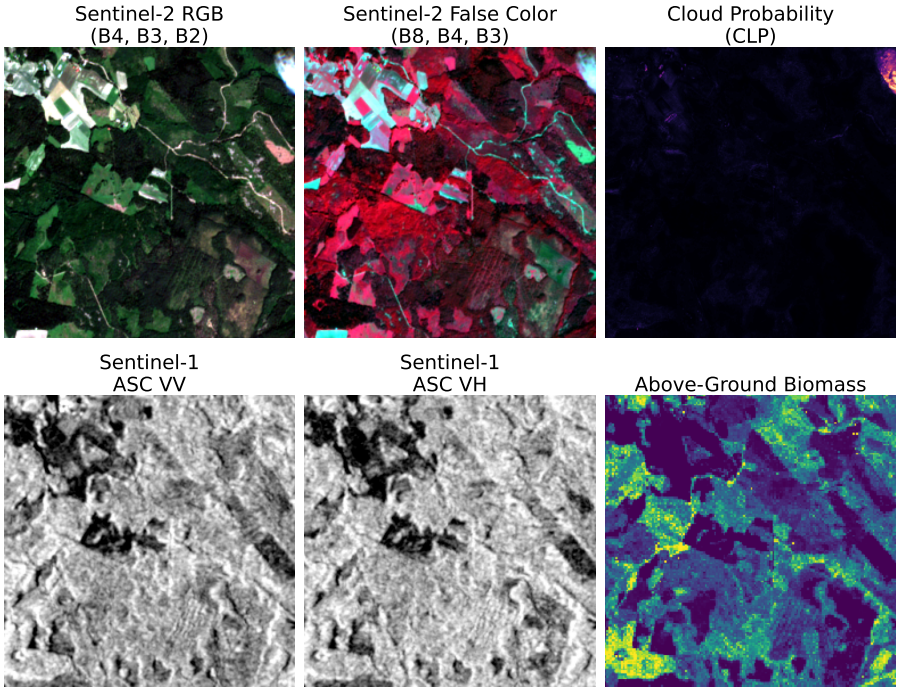


Figure 2: Example of the multisensor data provided for one sample. The top row shows Sentinel-2 RGB and false-color imagery together with the cloud probability layer, while the bottom row shows Sentinel-1 ASC VV and ASC VH observations and the corresponding reference above-ground biomass map. All images are spatially aligned.

Challenges.

- **Regression task.** Unlike semantic segmentation, where each pixel belongs to a discrete class, biomass estimation requires predicting a continuous value for every pixel. You will need to select an appropriate model architecture, loss function, and evaluation metrics for this dense regression problem.
- **Multimodal data.** Sentinel-1 and Sentinel-2 measure different physical properties of the Earth's surface. The two modalities have different statistical characteristics, and an appropriate strategy is required to combine their information effectively.
- **Input contribution analysis.** Sentinel-1 and Sentinel-2 provide multiple channels that capture different physical properties of vegetation and the Earth's surface. Beyond evaluating predictive performance, you will investigate which sensors and input bands contribute most to biomass estimation. You will need to design an appropriate analysis strategy and critically assess whether the observed contributions are consistent with the physical information captured by the different channels.

Expected outcome.

- Develop and compare regression models to estimate above-ground biomass from: 1) Sentinel-1 imagery alone, 2) Sentinel-2 imagery alone, and 3) combined Sentinel-1 and Sentinel-2 imagery. Explore different modeling strategies where appropriate.
- Evaluate the different models using relevant metrics and analyze their performance on the test set.
- Investigate the contribution of individual input bands or meaningful groups of bands to biomass estimation. Select and justify an appropriate methodology for assessing their importance. Possible approaches include ablation or perturbation experiments, permutation-based methods, or explainable AI (XAI) techniques.
- Relate the results of the contribution analysis to the physical information captured by the different Sentinel-1 and Sentinel-2 channels. Discuss sensor complementarity, limitations, and possible failure modes of the proposed models.

References

1 Nascetti, A., Yadav, R., Brodt, K., Qu, Q., Fan, H., Shendryk, Y., Shah, I., & Chung, C. (2023). BioMassters: A benchmark dataset for forest biomass estimation using multi-modal satellite time-series. Advances in Neural Information Processing Systems (NeurIPS)

3 Reconstructing cloud gaps in satellite ocean color with multi-source deep learning

Task. Surface chlorophyll-*a* (Chl-*a*) concentration retrieved from ocean colour satellites is a proxy for phytoplankton biomass, the base of the marine food web and a component of the ocean's uptake of carbon. Its retrieval relies on optical sensors, so every cloudy pixel is a missing value. On a daily global product, a large fraction of the ocean surface is unobserved on any given day, and the gaps are large and localized (Hong et al., 2022 [1]). This limits the use of ocean colour for monitoring blooms and upwelling events.
In this project, you will train a convolutional network to reconstruct Chl-*a* beneath clouds. The network sees the surrounding cloud-free pixels, a window of the days before and after, and, in addition, the gap-free field of Sea Surface Temperature (SST), provided by combining infrared, microwave, and in situ measurements. SST affects algal biological processes (e.g., metabolic rates, growth cycles,...) and carries the signature of the ocean circulation (upwelling, eddies and fronts) that advects the phytoplankton and shapes their distribution (Figure 2). Whether SST actually informs the reconstruction is a central question of the project (Ollier et al., 2026 [2]). Since no Chl-*a* measurement exists beneath a cloud, there is no ground truth to train against. A solution is to apply the cloud mask of a donor day (artificial cloud) to the input Chl-*a* at day t, and compute the loss only on the pixels that were valid in the original and hidden by the artificial cloud (Bachmann et al., 2022 [3]).
The study region is the Canary Current upwelling off Senegal and Mauritania coasts (21°W–15.7°W, 14°N–24°N). These waters sustain some of the most productive fisheries of the Atlantic, so the phytoplankton biomass has direct socio-economic relevance.

Data. The dataset is derived from two ESA Climate Change Initiative (CCI) products, already cropped to the study region and co-registered on a common grid. You can download it here.

- **Chl-*a*** : Ocean Color-CCI multi-sensor Level-3 daily product (`OCEANCOLOUR_GLO_BGC_L3_MY_009_103`) at 4 km resolution. Each day is a 240×127 pixels image. Pixels where the retrieval failed (essentially clouds) are set to `NaN`. This field is both an input and the prediction target.
- **Validity mask** : binary, 1 where Chl-*a* was retrieved and 0 where it was not. Derived from the field above and provided for convenience.
- **SST** : ESA SST CCI v3 Level-4 analysis (`SST_GLO_SST_L4_REP_OBSERVATIONS_010_024`), a gap-free daily product at 0.05° resolution. It is already bilinearly interpolated onto the Chl-*a* grid, so that the two fields can be stacked as channels without further regridding.

Both fields are daily and cover 2010-2020 (4018 days). The dataset can be split in time, e.g., 2010-2019 for training, 2020 for testing, with one year of the training period held out for validation. The split is by whole years on purpose, as Chl-*a* on consecutive days is strongly correlated, so a random split over individual days would place near-duplicate samples on both sides and inflate your scores.

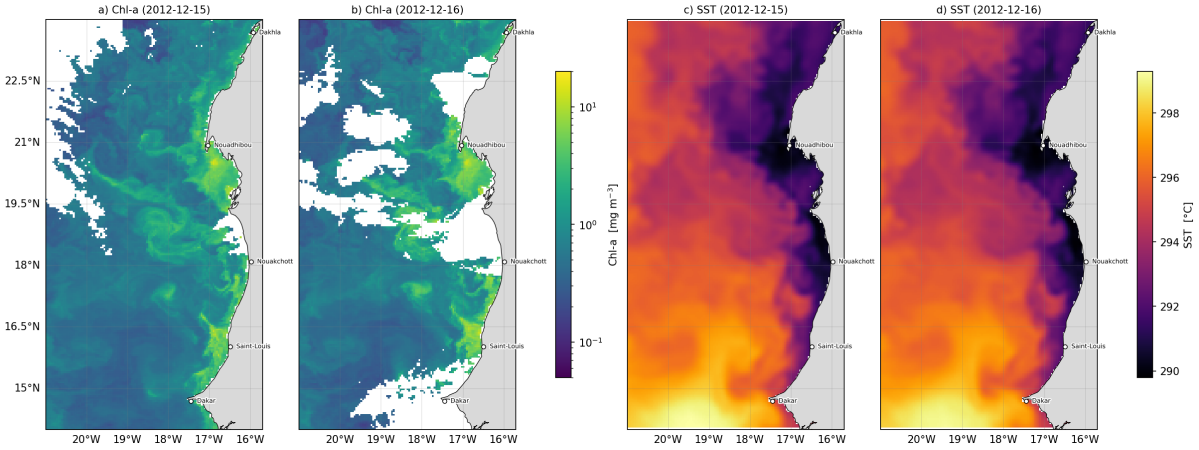


Figure 3: Example map for two consecutive days, the 15th and 16th of December 2012. a,b) Chl-*a* map displayed with a log-transform, because the chl-*a* distribution is skewed towards low values. c,d) Sea Surface Temperature (SST) map in degrees Celsius.

Expected Outcome.

- Build a training pipeline for a convolutional neural network that predicts Chl-*a* at every pixel. This is a regression task. The inputs can include Chl-*a* (days t-k ... t+k (k = 3 means a 7-day window), Chl-*a* validity mask, SST, and the output is Chl-*a* at day t. Work on $\log_{10}$(Chl-*a*) rather than raw concentrations, and NaN set at 0. Report RMSE and bias in $\log_{10}$ space, stratified by season, together with maps of the reconstructions and of their errors. Discuss the results and eventually your choices of deep learning architecture.
- Compare the results for three different input setups :
  - Chl-*a* + SST (with artificial masking of Chl-*a*, compute the loss on masked pixels only)
  - Chl-*a* alone (with artificial masking of Chl-*a*, compute the loss on masked pixels only)
  - SST alone (no artificial masking needed)
- Compare your model against two baselines that are not deep learning, for instance a per-pixel day-of-year climatology and a spatial interpolation of the valid pixels of the same day.

References

1 Hong, Z., Long, D., Li, X., Wang, Y., Zhang, J., Hamouda, M. A., & Mohamed, M. M. (2023). A global daily gap-filled chlorophyll-a dataset in open oceans during 2001–2021 from multisource information using convolutional neural networks. Earth System Science Data, 15(12), 5281-5300.
2 Ollier, L., El Hourany, R., Thiria, S., Mejia, C. E., Beaugrand, G., & Levy, M. (2026). Success and limits of reconstructing phytoplankton pigment temporal variations from simultaneous surface ocean physics with a Unet model. Machine Learning: Earth.
3 Bachmann, R., Mizrahi, D., Atanov, A., & Zamir, A. (2022, October). Multimae: Multi-modal multi-task masked autoencoders. In European conference on computer vision (pp. 348-367). Cham: Springer Nature Switzerland.

4 Spatial and temporal distribution shifts in wildfire forecasting

Task. In recent decades, wildfires have increasingly deviated from their natural role in Earth's ecosystems because of climate change and intensifying fire weather. As a result, we must better understand wildfire drivers and improve our ability to forecast wildfire risk accurately. Global wildfire forecasting models are commonly trained on historical fire observations from environmental and societal predictors. The geographical and temporal distribution of the training data may impact how well these models generalize to emerging wildfire patterns caused by climate change. In particular, an over-representation of Africa in the training dataset can affect predictions in new at-risk regions, such as North America, especially Canada.

In this project, you will develop a deep learning model for wildfire forecasting at an 8-day horizon: predictors at time t are used to predict burned pixels at $t+1$ to $t+8$, using the SeasFire dataset, and investigate how the composition of the training dataset impacts model performance in one or both directions:

- **Spatial analysis:** Investigate how the over-representation of African fires affects forecasting performance in North America, especially Canada.
- **Temporal analysis:** Investigate how the choice of training dataset affects forecasting performance, when constrained to one individual or a small subset of training years up to 3.

Data. The project will use a down-sampled version of the SeasFire dataset [1], a multivariate dataset for global wildfire forecasting covering 2002-2019 with the following temporal splits: train 2002-2017, validation 2018, and test 2019. SeasFire down-sampled provides weather variables, vegetation information, and human-related factors at a spatial resolution of $0.1°$ and a temporal resolution of 8 days; see Figure 4. The project will focus on forecasting burned patches at a horizon of 8 days (one interval step) with a subset of the predictors. The data cube is available to download here, and the partially preprocessed data is available here. A continent mask is also provided in the zip to define Africa and North America.

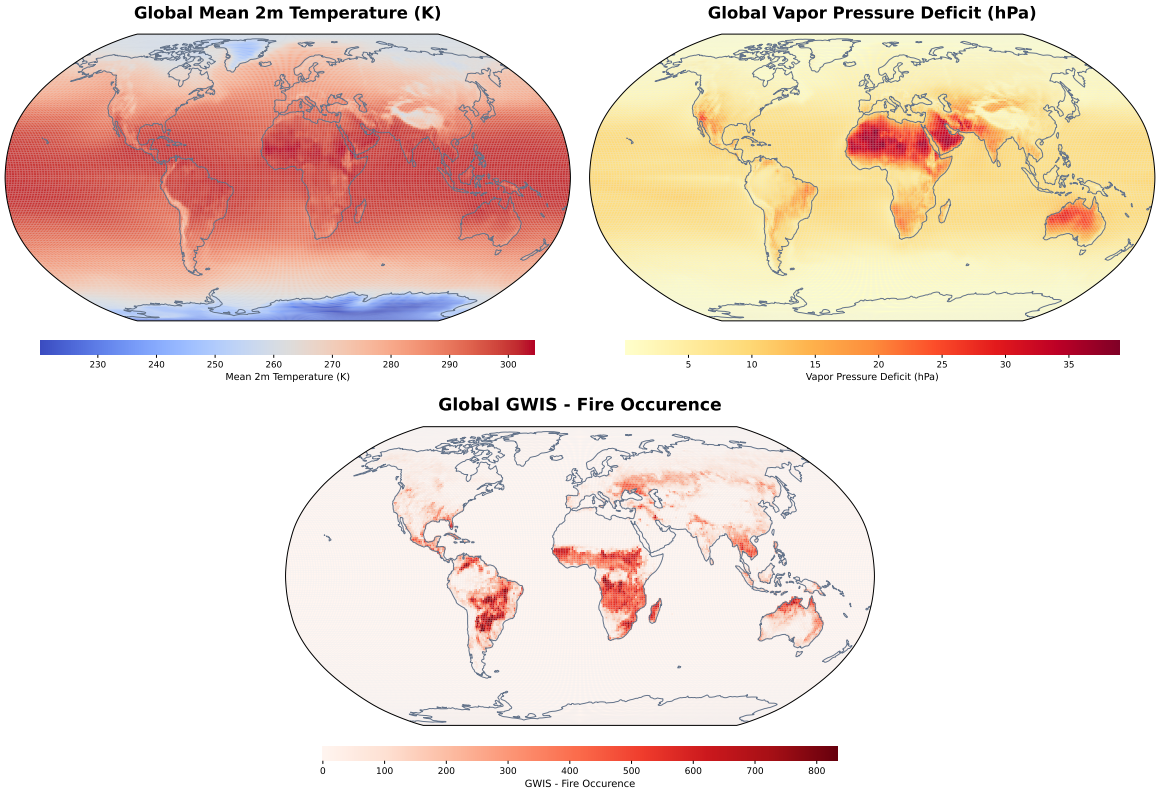


Figure 4: Representation of the down-sampled SeasFire dataset with two environmental predictors averaged over 2002-2019 and the total sum of per-pixel fire occurrences over the same period.

Expected Outcome.

- Implement a training pipeline for a dense deep learning model to predict burned pixels from environmental and societal predictors.
- Evaluate model performance using metrics robust to potential class imbalance on the test year, globally and per region.
- Explore one or both of the following directions:
  - Compare the original model performance on North America with training excluding Africa and with training where the same amount of data is removed at random, to separate the effect of removing Africa from the effect of having less data. Investigate potential strategies for geographically balanced training, to optimize evaluation metrics over North America.
  - Compare models trained on different individual years or a small subset of years, and investigate potential strategies for the selection of training years using the similarity to the target period, computed from predictors only. Compare this strategy against a fixed or arbitrary training-year baseline (e.g. most recent years, randomly chosen years) and against training on all years.

Developing one direction is enough for full points; the second one will be considered as a bonus.

Challenges.

- **Dense prediction task:** Unlike standard image classification tasks, here each pixel needs to be classified as fire or no-fire class. Moreover, the dataset is fairly imbalanced towards the no-fire class.
- **Training-year selection:** The selection strategy must be evaluated on a held-out year without using target-period wildfire labels. Design your strategy on the validation year (2018), then apply it unchanged to the test year (2019).
- **Experimental design:** Consistent train, validation, and test splits are required to compare the different strategies fairly. Training sets of different sizes should be compared with the same model and training budget (number of iterations).

References. [1] Karasante, Ilektra, et al. ”SeasFire cube-a multivariate dataset for global wildfire modeling.” Scientific Data 12.1 (2025): 368.

5 Reconstructing land surface temperature from satellite images

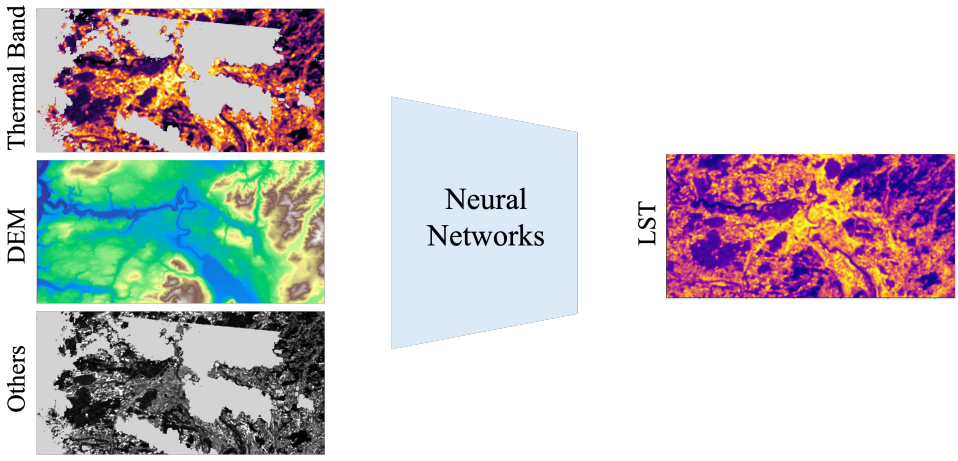


Figure 5: An overview of the LST reconstruction process. Different modalities are provided as the input to the model to predict LST and reconstruct missing regions.

Task. Land Surface Temperature (LST) is a key variable for applications including agricultural management and urban planning. Estimating land surface temperature using numerical methods tends to be computationally intensive and requires complex assimilation processes. Deep learning provides a way to efficiently predict LST from Earth observation sensors. However, the heterogeneity of sensors and incompleteness of observations (e.g., cloud obstruction) pose challenges to reconstructing high-resolution LST. In this project, you will develop deep learning methods that integrate different data sources, including satellite imagery and digital elevation models, to predict high-resolution LST and reconstruct missing regions.

Data. The dataset is available under the following link. We provide 4 relevant bands (SR_B4, SR_B5, BT_B10, BT_B11) derived from Landsat-8 [1] (see input folders) and the Digital Elevation Model of a 30.72×15.36 km region centered on Bern (bern_dem_swissalti3d_30m.tif) as the observational data. As the target, we provide the LST product (see target folders) paired with each Landsat-8 image. All the data has been gridded to 1024×512 pixels on a 30 m resolution grid. We split the dataset into 147 training pairs with inputs and targets with the same obscured regions, 72 test-task1 pairs with inputs and targets with the same obscured regions, and 28 test-task2 pairs with obscured inputs and complete LST targets.

Expected Outcome.

- Develop models to predict incomplete LST given input modalities with incomplete observations of the same obscured regions, then validate the model on test-task1.
- Develop models to reconstruct complete LST given input modalities with incomplete observations, then validate the model on test-task2. Explore one of the following directions:
  - Develop data-driven models that learn to reconstruct missing areas with self-supervised learning objectives [2].
  - Develop hybrid models that leverage relationships between the complete reference modality (DEM) and the target variable to reconstruct the missing area [3].
- Explore the effects of different modalities on the performance of predicting the LST. Use tools to visualize and analyze the results.

Challenges.

- **Computational overhead.** Processing high-resolution images from different Earth observation sensors can introduce significant GPU memory consumption. Preprocessing methods (e.g., tiling) may be required to reduce GPU consumption when loading images.

- **Incomplete observations.** Earth observation imagery often contains missing regions, for example due to cloud cover. These regions require explicit handling during land surface temperature (LST) prediction. A review and discussion are also needed to assess how LST can be estimated in these regions from incomplete observations.

References

1 Earth Resources Observation and Science (EROS) Center. Landsat 8-9 Operational Land Imager / Thermal Infrared Sensor Level-2, Collection 2. U.S. Geological Survey, 2020.
2 He, Kaiming, et al. "Masked autoencoders are scalable vision learners." 2022 IEEE/CVF conference on computer vision and pattern recognition (CVPR). IEEE, 2022.
3 Zgraggen Jan, et al. "Topography-aware Temperature Forecasting in Europe", Climate informatics, 2026, link.